

Identification of Automated Music Genre by Exploring Machine Learning Approaches

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Abstract— Music genres are some groups of terms generated for the sake of categories the music. Music genres have some typical characteristics. The characteristics are associated with harmonic content of the music, rhythmic structure, and instrumentation. Basically, these Genre hierarchies are utilized to arrange the big collection of songs on the internet. At present this is done manually, so automatic classification is necessary in order to aid the human user. Also, automatic genre classification for music imparts an outline for evaluating and developing attributes for the interpretation of musical signals. For classification purpose different genre classes such as Blues, classical, pop, disco, hip-hop, jazz, reggae, country, metal, and rock. In this proposed work four widely known machine learning algorithms are used to train and test the classifier with a well-known dataset for audio data classification to envisage the genre of music that the audio given as input belongs to. Convolution Neural Network(CNN), Feed-Forward Neural Network(FNN), K- Nearest Neighbours(KNN), Support Vector Machine(SVM) classifiers are used out of which Convolution Neural Network achieved highest accuracy of 81%.

Keywords— Music genres, convolution neural networks, classification, Feed-Forward Neural Network, K- Nearest Neighbours, Support Vector Machine

I. INTRODUCTION

In today's world, there is a huge rise in online music streaming platforms, where people can sit anywhere and listen to music at any point in time. This widespread of online music platforms has led to hurdles related to music copyright. Today almost everyone is making use of the digital realm to listen to online music through various platforms such as Jio Savan, Wynk Music and so on. With the uprise in the number of online music platforms and ease of access, it has become difficult to manage them. Songs can be categorized based on genre which depends on different characteristics. Audio feature extraction is an essential task of audio synthesis, effect design, processing, and music information retrieval [1]. The musical context of the songs can be described using tags generated by the music genre which is presented in this paper. In order to achieve satisfactory performance, audio features must be extracted. Long-term audio features are acquired by approximating the statistics of short-term features or sometimes it is based on audio-feature descriptions of audio signals. Hence Conventional bag-of frame approaches have been proposed.

Different music streaming applications are used to listen to music by almost everyone. As per today the leading one is Spotify with a net worth of \$26 billion. This application has

millions of songs and entitlements to search for the right song for all. The study says Spotify has invested a tremendous amount of research to expand this way. Spotify used recent advanced technology like machine learning, Natural language Processing (NLP) [2], collaborative filtering, deep learning and many more. Songs are examined based on tempo, dance ability, acoustics, energy, etc. Machine learning methods are completely successful in mining large pools of data. This work needs some knowledge like handling sound, compute sound and audio features from them and finally apply Machine Learning Algorithms on them. The aspiration of this system is to automate the music classification making the selection of the song rapid. If anybody is given a chance to categorise songs it can be done by listening to whole music which is a time-consuming and difficult task. Hence automating music classification would provide valuable data like trends, popular gene, artists without difficulty. In this study CNN, FNN, KNN, SVM classifiers were used out of which CNN achieved highest accuracy of 81%.

The paper follows a logical structure, starting with an introduction in Section I, Related works in the field of classification of music genre in Section II, followed by the methodology in Section III, Result analysis in Section IV and finally, concluding with the achievements and future directions.

II. RELATED WORKS

Hareesh Bahuleyan [3] used both machine learning and neural networks for music genre classification by attaching tags to the songs available in the library. Firstly, CNN is trained using different features of Spectrograms of the audio signals. Secondly different machine learning methods Random forest, Logistic regression are used. When the manually extracted features like Spectral Centroid, Chroma Features, Mel-Frequency Cepstral Coefficients (MFCC) are obtained, it is used to classify by considering machine learning algorithms. The ensemble model obtained an accuracy of 89%. An overall accuracy of 61% is achieved for 10 musical genres. Athulya et. al [4] used deep learning techniques to classify the music genre using spectrograms. Also, a recommendation system is implemented which recommends songs based on user's preference. The experiments were carried out using the GTZAN dataset.

Anirudh et.al. [5] mentioned in the passage focuses on comparing different classification models and proposing a

new model reliant on convolutional neural networks (CNN). The CNN model proposed in this research is claimed to outperform previously proposed models. The evaluation and comparison of the models were conducted on the GTZAN dataset, which is a widely used benchmark dataset for music genre classification tasks. By leveraging machine learning techniques and specifically CNNs, the research aims to enhance the accuracy and efficiency of music genre classification. This automated approach can significantly contribute to organizing and satisfying the categorical needs of music listeners in an ever-expanding market with diverse music styles.

The study by Ahmet Elbir [6] focuses on music genre prediction in the field of digital music processing. The researchers extract acoustic features from music using digital signal processing techniques. They then employ machine learning methods for music genre classification and music

recommendations based on these extracted features. In particular, the study explores the use of convolutional neural networks (CNNs), which are deep learning methods, for genre classification and music recommendation tasks. The researchers compare the performance of distinct algorithms and techniques, and it is mentioned that the ultimate success rate was achieved using the Support Vector Machine (SVM) algorithm.

The CNN architecture proposed by Califeng Liu et. al. [7] in the paper considers long contextual information, allowing for the transfer of more appropriate information to the decision-making layer. The paper reports conducting various experiments on benchmark datasets such as GTZAN, Ballroom, and Extended Ballroom to estimate the performance of the proposed neural network. The results illustrate excellent performances of the proposed CNN architecture in music genre classification tasks.

TABLE I. OUTLINE OF RELATED WORKS

Source	Dataset used	Results and Discussion
Hareesh Bahuleyan [3]	Sound clips of YouTube Videos	Ensembling the CNN and XGBoost model could lead to accuracy of 85%
Athulya et. al. [4]	GTZAN	Using spectrogram feature values genre classification was done with accuracy of 94%
Anirudh et.al.[5]	GTZAN	Multiple models were produced. Decision tree achieves 91% of accuracy.
Ahmet Elbir [6]	GTZAN	Classification and recommendation of songs were done using acoustic features. SVM achieved higher accuracy which was even better than CNN.
Califeng Liu et. al. [7]	GTZAN, Ballroom, and Extended Ballroom	Broadcast Neural Network (BBNN) gives better accuracy of 90%
Lin Feng et. al. [8]	GTZAN	Hybrid architecture with paralleling CNN and Bi-RNN could achieve 92% of Accuracy
Nagamanoj Karunakaran et. al. [9]	Free Music Archive (FMA) and GTZAN Dataset	Blurry Classification problem is overcome using Hybrid classifier. Overall accuracy is 90% and 70% resp.

Table I shows outline of related works. Music genres can be subjective and have overlapping characteristics, making them difficult to define precisely. Music genre datasets often suffer from class imbalance, where some genres have a significantly larger number of samples compared to others. This imbalance can lead to biased models that perform well on dominant genres but struggle with underrepresented ones. Handling cross-genre variability is crucial for accurate and robust genre classification. Selecting appropriate audio features that effectively capture genre-related information is a non-trivial task. Different features may work better for certain genres, and finding a balance between discriminative and generalizable features is essential. In the proposed work different classifiers are used to classify the music genre more effectively.

III. METHODOLOGY

Fig. 1 shows the block diagram of proposed model. The audio that is recorded for generic classification is processed so that it can be classified into its respective genre. In order to understand that audio signals should be visualized, mainly audio signals can be visualised using two ways, they are Time series visualisation, Spectrogram visualisation. In this project Spectrogram Visualisation is used for audio data processing.

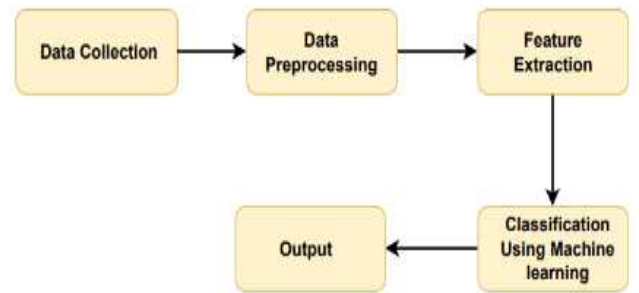


Fig. 1. Block diagram of proposed model

A. Data Collection

GTZAN [10] music genre collection dataset is used which consists of a thousand songs. This dataset is split into two sets separating it as 80:20 for training and testing respectively. These songs are of 30 seconds duration. It has ten genres like classical, blues, country, rock, disco, reggae, hip-hop, jazz, metal, and pop. Pre-emphasis filter is useful in order to boost amplitudes at high frequencies.

B. Feature Extraction

Feature Extraction is one of essential task. Features are extracted using Mel-spectrograms, Spec- centroid, Spectral

roll-off, Zero-crossing rate, Spectral bandwidth, and Chromo frequencies. Feature extraction plays a wide role in analysing and identifying the relations between songs and genres. Once the feature extraction is done data augmentation is used in order to maximize the training data. By this accuracy of the model can be maximized by reducing the overfitting augment function, horizontal flip and the random cut out are used to improve the size of the training data. Horizontal flip is used to flipping the spectrogram feature NumPy array and create a new input. In random cutout, a random NumPy array is generated and dB values are filled in the places to make it an actual spectrogram signal in the frequency domain. These features were extracted using librosa.

C. Classification using Machine Learning

Classification using machine learning [11] is a popular and powerful technique for categorizing data into predefined classes or categories. It is a supervised learning task where a model learns patterns from labelled training data and uses that knowledge to predict the class labels of unseen or future data points. Convolutional Neural Networks [12-13] are a type of deep learning model commonly used for image recognition tasks. However, they can also be applied to other types of data, including audio signals such as music, for tasks like music genre identification. In the context of automated music genre identification using CNNs, the input to the network is typically a spectrogram representation of an audio clip. A spectrogram is a visual representation of the spectrum of frequencies of a signal as it varies with time. It can be obtained by applying a Fourier transform to short segments of the audio signal over time.

A Feed Forward Neural Network [14], also known as a multilayer perceptron, is a type of artificial neural network commonly used for various machine learning tasks, including music genre identification. FFNNs are designed to process input data through a series of interconnected layers, where information flows in a forward direction, from the input layer to the output layer, without any cycles or loops. In the context of automated music genre identification, an FFNN takes audio features extracted from music as input and predicts the genre label associated with the input. These audio features can include various representations of the music signal, such as spectrograms, mel-frequency cepstral coefficients (MFCCs), or other engineered features.

K-Nearest Neighbours (KNN) [15-16] is a simple yet effective algorithm used for classification tasks, including automated music genre identification. KNN works by finding the k nearest neighbors in the training dataset to a given test instance and assigning the most common class label among those neighbours as the predicted label for the test instance. presents an application of KNN in music genre classification using Mel-frequency cepstral coefficients (MFCCs) as audio features.

Support Vector Machine is a popular machine learning algorithm used for classification tasks, including automated music genre identification. SVMs are particularly effective in handling high-dimensional data and finding optimal decision boundaries between different classes. D. R. Ignatius Moses Setiadi [17-18] investigated the effect of feature selection on the accuracy of music genre classification using a Support Vector Machine (SVM) classifier. Feature selection is a process of selecting a subset of relevant features from the original set of features to improve the classification performance and reduce computational complexity.

IV. RESULT ANALYSIS

Since the audio is recorded from external sources the audio signal generated by the machine learning classifiers are never accurate for all frames of the entire music. It was found that by combination of all machine learning algorithms an accuracy of 81% was obtained. Machine learning techniques but the overall accuracy singles handily depended on CNN is a powerful machine learning model when it comes to audio data processing and classification. From the confusion matrix it shows that the “rock” genre as it is predicted accuracy is the lowest standing at 0.32. Our model does a pretty good job in classifying music into genres such as disco, hip-hop, pop and classical with predictive accuracy of above 70% with the highest true positive classification being up to 99%. Mel Spectrogram generation for a maximum frequency of 8Khz, for a sampling rate of the Blues music genre shown in Fig 2.

Python programming language is used for the classification of music genre. Some of the machine learning libraries like sci-kit-learn, Keras used for GUI purpose. Matplotlib is used to create visualizations of the extracted features and classify them. The set of short-term power spectrum characteristics of the sound is represented by Mel-Frequency Spectral Coefficients and have been used in the state-of-the-art recognition and sound categorisation techniques, it models the characterises of human voice. Formula to work with Mel Scale is shown in equation 1.

$$M(f) = 1125 \ln(1 + f/700) \quad (1)$$

Spectral Band-width: The p-th order spectral band-width corresponds to the p-th order moment about the spectral centroid and is calculated as followed. Visualization of Spectral centroid for audio signal is shown in Fig. 3. Spectral Bandwidth for $2 < p < 4$ and audio with sr=24000 signal is depicted in Fig.4. and Fig. 5. Table II shows different performance metrics like Precision, Recall, F1-score, and support for different genre groups like G0 to G9. Table II and Fig 6 shows the classification metric of genre class where G0= blues, G1=classical, G2=country, G3=disco, G4=hip hop, G5=jazz, G6=metal, G7=pop, G8=reggae, G9=rock. By comparing the performance of each genre G9 class performance is lower and the model finds it difficult to identify the G9 genre. Figure 7 shows the accuracy of the model over 500 epochs and an accuracy of 81% is achieved.

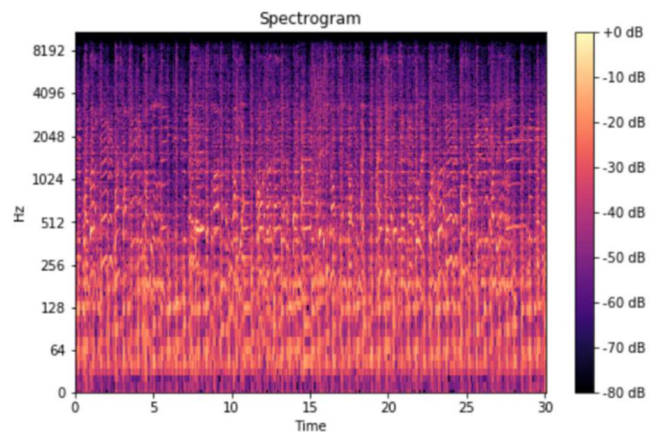


Fig 2. Mel Spectrogram for blues genre

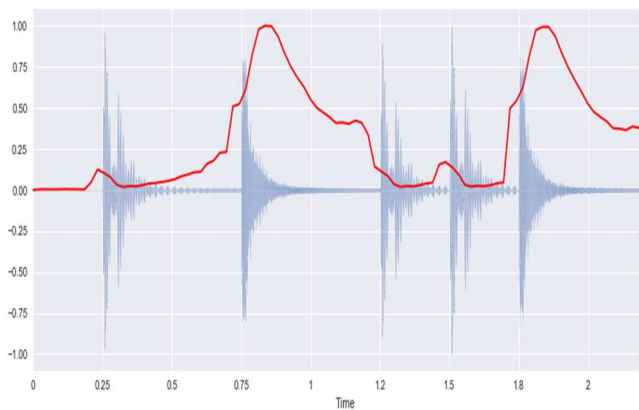


Fig. 3. Visualization of Spectral centroid for an audio signal.

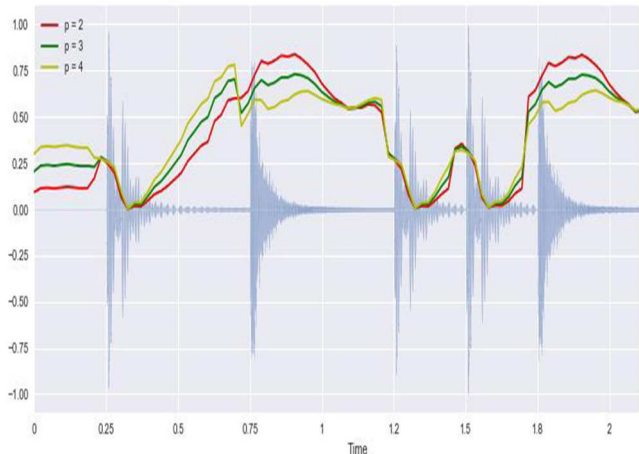


Fig. 4. Spectral Bandwidth for $2 < p < 4$

For CNN Model. Figure 8 shows the CNN model loss measure where loss measure is analysed over epochs of 500 and loss decreases gradually and yields better results. Figure 9 and 10 shows the CNN Model Training Loss and accuracy over the epochs of 80. Training accuracy of CNN model has achieved better score.

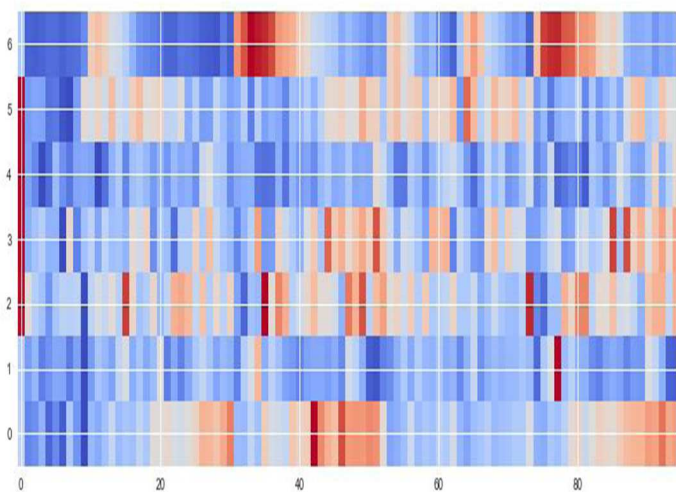


Fig. 5. Spectral contrast for the same audio with $sr=24000$ signal

TABLE II. PERFORMANCE RESULTS FOR GENRE CLASSES

Genre Class	Precision	Recall	F1 score	Support
G0	0.90	0.88	0.89	390
G1	0.96	0.99	0.97	390
G2	0.92	0.94	0.93	390
G3	0.91	0.94	0.93	390
G4	0.98	0.98	0.98	390
G5	0.94	0.97	0.95	390
G6	0.94	0.90	0.92	390
G7	0.99	0.94	0.96	390
G8	0.97	0.94	0.95	390
G9	0.81	0.83	0.82	390
Accuracy			0.94	3900
Macro Average	0.94	0.94	0.94	3900
Weighted Average	0.94	0.94	0.95	3900

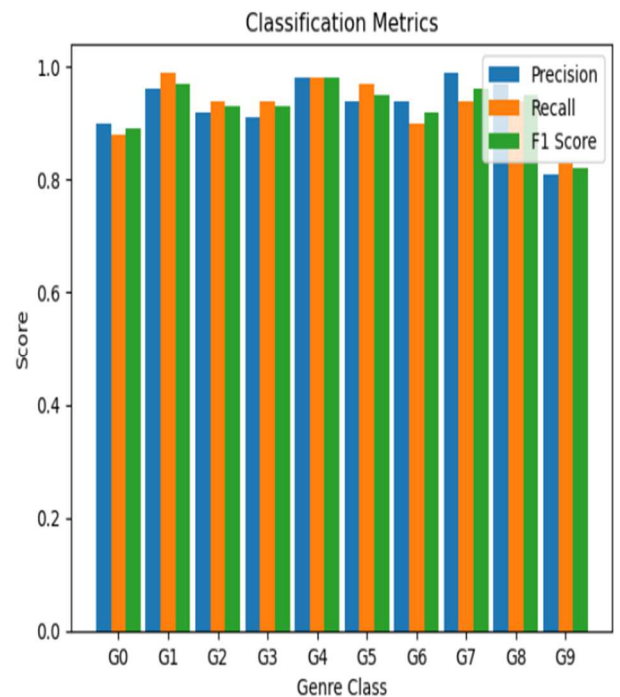


Fig 6: Classification Metric of Genre Classes

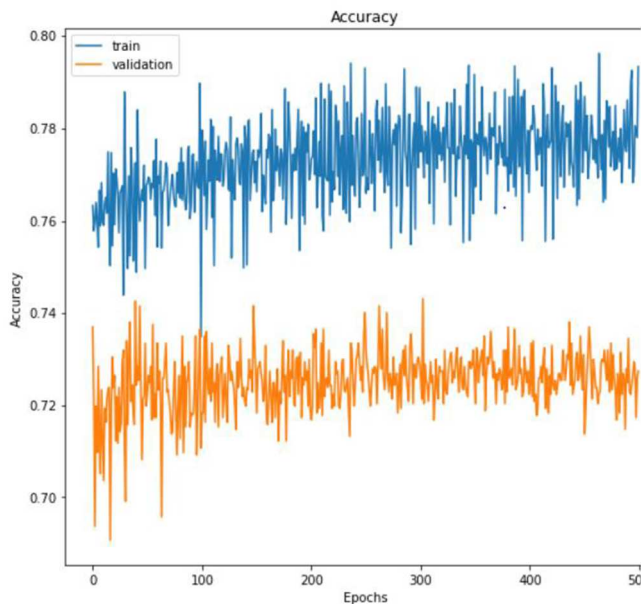


Fig. 7. CNN Model Accuracy Measure

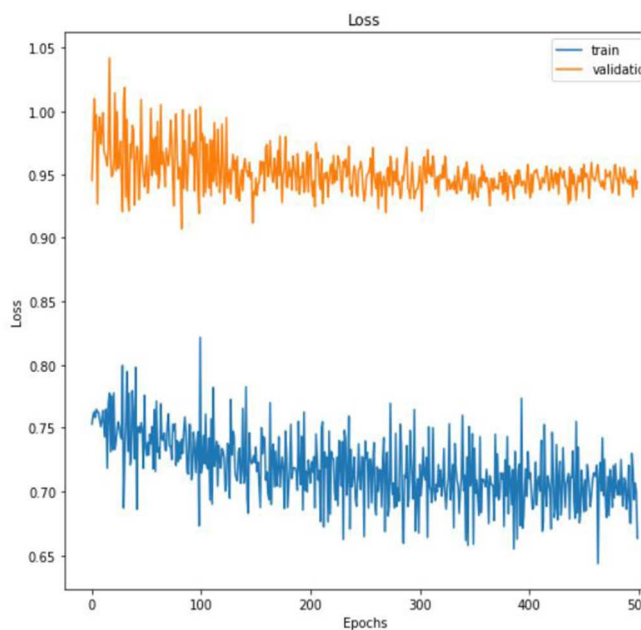


Fig.8. CNN Model Loss Measure

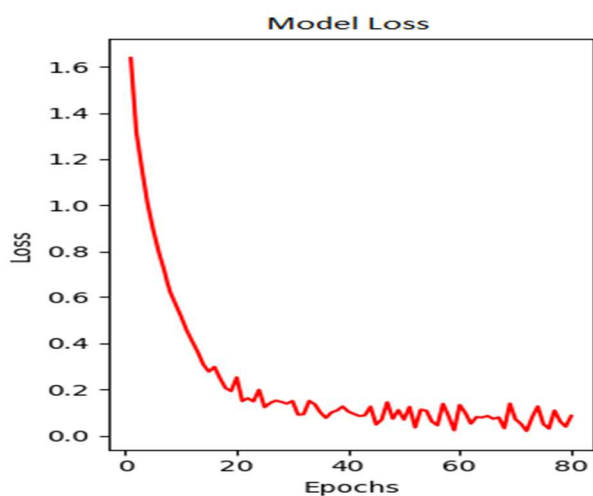


Fig. 9. CNN Model Training Loss

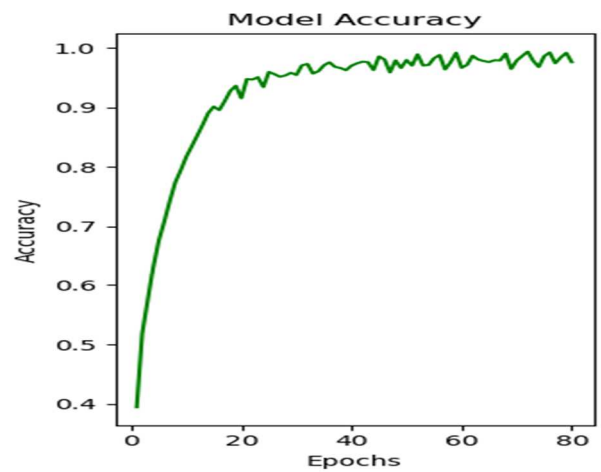


Fig 10: CNN Model Training Accuracy

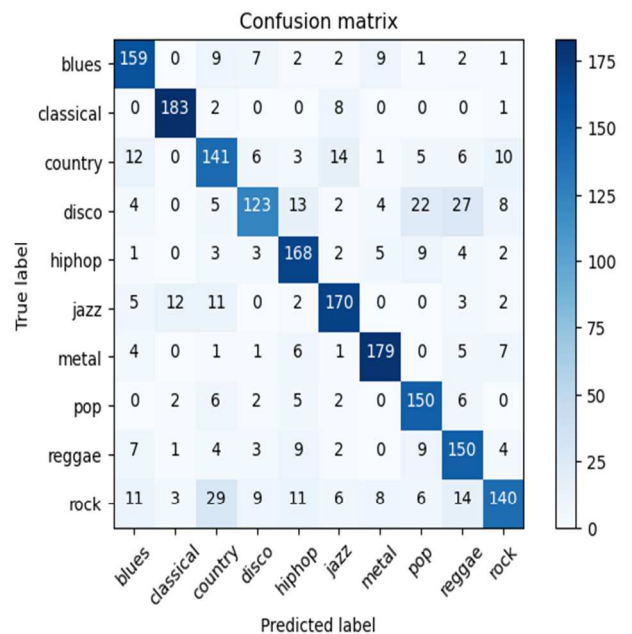


Fig 11: Confusion Matrix of CNN Model

Fig 11 shows the Confusion Matrix of the CNN model on the test data where 10 classes have been considered such as blues, classical, country, disco, hip hop, jazz, metal, pop, reggae, rock. Confusion matrix compares the prediction value against actual value.

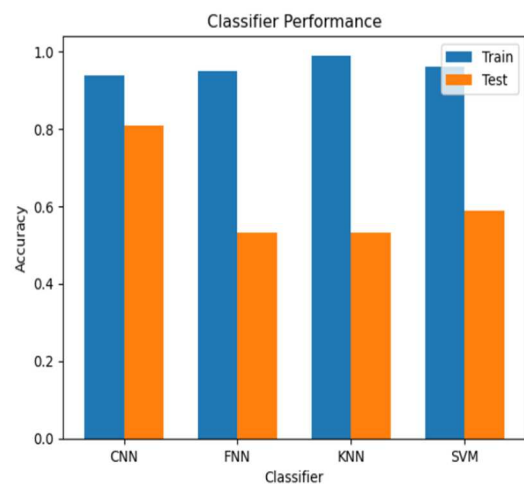


Fig 12: Classification Performance of Algorithms

Fig 12 and Table III shows the classification accuracy of the different algorithms CNN model has achieved better accuracy compare to other algorithms. Ratio of 70:30 where 70% data for training and 30% data for testing is considered for evaluating the model performance.

TABLE III. ACCURACY OF THE MODEL

Classifier	Train	Test
Convolution Neural Network(CNN)	0.94	0.81
Feed-Forward Neural Network(FNN)	0.95	0.53
K- Nearest Neighbours(KNN)	0.99	0.53
Support Vector Machine(SVM)	0.96	0.59

V. CONCLUSION AND FUTURE WORK

The identification of automated music genre using machine learning approaches has witnessed significant advancements and research efforts. Different algorithms, such as CNN, SVM, KNN, and FNN, have been applied to this task, each with its own strengths and limitations. CNN have demonstrated their effectiveness in capturing hierarchical and local features from music data, particularly when combined with spectrogram-based representations. SVMs excel in handling high-dimensional data and finding optimal decision boundaries, while KNN offers a simple and intuitive approach that relies on similarity measures. FNN provide versatility and can handle both linear and non-linear relationships in the data. CNN achieved highest accuracy of 81%.

Future work in the identification of automated music genre could explore several directions:

- Hybrid models: Investigating the combination of multiple machines learning algorithms or incorporating ensemble methods
- Advanced feature representations: Deep learning-based feature learning to capture more intricate characteristics of music for better genre identification.
- Handling unbalanced datasets: Strategies to address class imbalance issues that often exist in music genre datasets, such as oversampling minority classes.
- Real-time classification: Developing efficient and lightweight models suitable for real-time music genre identification applications, such as music streaming platforms or automatic playlist generation systems.

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